

Erdős Problem #782

EP-782

L1 - TRACTABLE

PARTIALLY SOLVED

Combinatorics

ORIGINAL PROBLEM STATEMENT

Do the squares contain arbitrarily long quasi-progressions? That is, does there exist some constant $C > 0$ such that, for any k , the squares contain a sequence $x_1, \dots, x_k$ where, for some d and all $1 \leq i < k$, $x_i + d \leq x_{i+1} \leq x_i + d + C$. Do the squares contain arbitrarily large cubes $a + \{\sum_i \epsilon_i b_i : \epsilon_i \in \{0, 1\}\}$?

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>